

THERESA, Antarctica

WMO: 893140

Lat: 84.600S

Lon: 115.817W

Elev: 4800

StdP: 12.32

Time Zone: -8.00 (W08)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-51.9	-47.0	N/A	N/A	N/A	N/A	N/A	N/A	52.1	-21.5	46.6	-23.5	12.9	90	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	8.6	22.9	N/A	20.4	N/A	17.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	12.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 47.2	42.7	38.7	DB	-57.8	30.8	4.5	4.9	-61.1	34.3	-63.7	37.2	-66.3	39.9	-69.6	43.5	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-8.7	10.8	0.1	-9.3	-14.4	-14.8	-18.0	-21.5	-20.4	-18.7	-10.1	0.8	11.4	(6)	
(7)		DBStd	14.60	5.55	7.36	9.85	9.20	10.52	11.17	11.40	11.22	11.41	8.61	6.58	4.78	(7)	
(8)		HDD50	21425	1214	1398	1838	1931	2010	2040	2217	2182	2061	1863	1475	1196	(8)	
(9)		HDD65	26900	1679	1818	2303	2381	2475	2490	2682	2647	2511	2328	1925	1661	(9)	
(10)		CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(10)	
(11)		CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
(12)		CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)	
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14)	Wind	WSAvg	21.2	14.3	17.4	20.9	22.4	24.6	22.7	23.5	24.8	24.2	23.3	20.9	14.9	(14)	
(15)	Precipitation	PrecAvg	11.6	0.4	0.4	0.7	0.8	1.3	1.3	1.5	1.5	1.6	1.1	0.7	0.4	(15)	
(16)		PrecMax	17.8	1.1	0.9	1.4	2.0	2.4	2.4	3.8	2.9	3.0	2.4	2.2	1.1	(16)	
(17)		PrecMin	6.7	0.0	0.2	0.2	0.2	0.3	0.4	0.4	0.6	0.3	0.2	0.1	0.1	(17)	
(18)		PrecStd	2.6	0.3	0.2	0.4	0.5	0.5	0.5	0.7	0.7	0.7	0.6	0.5	0.2	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.0	17.9	14.9	9.3	13.8	9.7	6.7	6.5	10.0	11.1	18.3	26.7	(19)	
(20)			MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(20)
(21)		2%	DB	23.7	14.5	11.5	4.1	4.7	4.7	0.2	2.9	4.6	7.6	14.7	23.1	(21)	
(22)			MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(22)
(23)		5%	DB	21.0	12.5	8.2	0.3	1.1	0.6	-3.2	-2.0	0.5	4.7	12.2	20.7	(23)	
(24)			MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(24)
(25)		10%	DB	18.6	10.1	4.5	-2.3	-1.9	-2.8	-6.5	-5.5	-3.6	1.8	9.9	18.1	(25)	
(26)			MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(27)	
(28)			MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(28)
(29)		2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(29)
(30)			MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(30)
(31)		5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(31)
(32)			MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(32)
(33)		10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(33)
(34)			MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(34)
(35)	Mean Daily Temperature Range		MDBR	8.4	8.2	9.2	9.8	10.5	10.7	11.0	11.7	10.9	8.7	7.9	8.6	(35)	
(36)		5% DB	MCDBR	12.2	8.8	9.8	11.2	12.2	13.9	12.4	12.7	13.9	10.2	10.1	13.2	(36)	
(37)			MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(37)	
(38)			MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(38)
(39)		5% WB	MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.181	0.163	0.106	N/A	N/A	N/A	N/A	N/A	N/A	0.102	0.155	0.171	0.180	(40)	
(41)		taud	2.277	2.231	1.939	N/A	N/A	N/A	N/A	N/A	N/A	1.893	2.193	2.270	2.287	(41)	
(42)		Ebn at Noon	316	286	204	0	0	0	0	0	198	290	320	326	(42)		
(43)		Edh at Noon	25	18	8	0	0	0	0	0	8	19	25	27	(43)		
(44)	All-Sky Solar Radiation	RadAvg	2726	1487	275	0	0	0	0	0	83	1000	2438	3200	(44)		
(45)		RadStd	77	61	13	0	0	0	0	0	5	37	92	106	(45)		

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page